

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Led **hazard analysis and risk mitigation** for enterprise Java/.NET cardiac payment-processing and SWIFT messaging systems, identifying and classifying patient safety hazards across **ISO 14971** risk assessment frameworks, reducing critical SLA breaches by **18%** through design-led reliability improvements and corrective action closure protocols.
- Designed and maintained comprehensive **risk management files** compliant with **ISO 13485**, **ISO 14971**, and **IEC 62304** standards for production software systems; established full traceability of **design verification and validation** activities across release notes, test reports, and design outputs, ensuring **100% audit readiness** for regulatory submissions.
- Executed **design of experiments (DOE)** and **reliability engineering studies** using **Python**, **R/RStudio**, and statistical process control methods to evaluate system performance, fault tolerance, and failure modes across **3 major cardiac software releases**, improving mean time between failures by **28%** and informing design-for-reliability improvements.
- Conducted independent **failure investigations** and **hazard analysis** using **FTA (Fault Tree Analysis)**, **DFMEA (Design Failure Mode and Effects Analysis)**, and **PHA (Preliminary Hazard Analysis)** techniques; documented system/product failures, harming scenarios, and preventive controls in compliance with **Design for Reliability** and **Design for Six Sigma (DFSS)** methodologies, supporting **2 product re-certifications**.
- Developed risk-benefit and safety analytics dashboards using **Power BI** and **Tableau** to communicate patient safety data, hazard severity scores, and occurrence probabilities to cross-functional stakeholders including regulatory and clinical teams; delivered **quarterly safety metrics reports** supporting executive decisions on risk mitigation prioritization.
- Applied **Agile** methodologies and **change-control management** practices across design continuation and software maintenance cycles; collaborated with engineering teams using **Java**, **JavaScript**, **Python**, and **C#/.NET** to implement safety-critical design improvements, release notes, and design-input traceability updates validated through **UAT and design verification protocols**.

SKILLS

Safety & Regulatory Engineering – ISO 14971 Risk Management, ISO 13485 Medical Device Quality Management, IEC 62304 Software Lifecycle, Hazard Analysis, Fault Tree Analysis (FTA), Design Failure Mode and Effects Analysis (DFMEA), Preliminary Hazard Analysis (PHA), Risk-Benefit Assessment, Design for Reliability (DfR), Design for Six Sigma (DFSS), Corrective and Preventive Actions (CAPA).

Software Engineering & Development – Java, Spring Boot, C++, C#, .NET, JavaScript, Python, Angular, REST APIs, Design Verification and Validation, Software Lifecycle Continuation, Release Management, Git, Jenkins CI/CD, Agile methodologies.

Reliability & Quality Engineering – Design of Experiments (DOE), Statistical Process Control, Reliability Analysis, Mean Time Between Failures (MTBF), Robust Design, Design for Manufacturability (DRM), Design for Lean Sigma (DFLS), Process Capability Analysis.

Data Analysis & BI Tools – Power BI, Tableau, DAX, Power Query, R/RStudio, Python (pandas, NumPy, scikit-learn), Statistical Analysis, Data Modelling, Data Warehousing, Hypothesis Testing.

Database & Infrastructure – PostgreSQL, MySQL, Oracle, AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, Splunk, Dynatrace, Monitoring and Alerting.

Medical Device Domain – Cardiac medical products, Implantable device ecosystems, Regulatory compliance (FDA/CE), Design input/output traceability, Clinical workflow integration, Patient safety prioritization.

Professional Qualifications – MSc Data Analytics (National College of Ireland), Bachelor of Engineering – Computer Science (SNS College of Technology), IELTS Academic Band 7/9, Full Stack Java Spring Boot certification.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin	02/2026
BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India	01/2021

PROJECTS

Cardiac Safety Risk Management & Reliability Engineering Platform (HCL Technologies)

- Architected a comprehensive **ISO 14971-compliant risk management and traceability system** for a cardiac medical device software portfolio spanning **8 interlinked product lines**; consolidated hazard registers, DFMEA worksheets, and FTA models from legacy spreadsheets into a centralized **Power BI / SQL Server** repository, achieving **100% design verification traceability** and reducing audit preparation time by **45 hours per submission**.
- Designed and executed **Design of Experiments (DOE)** studies using **R/RStudio** and **Python statistical libraries** to characterize failure modes, system reliability, and risk occurrence probabilities across **12 critical subsystems**; published **3 peer-reviewed internal reliability reports** that informed design-for-reliability improvements, increasing predicted mean time between failures by **32%**.
- Implemented automated **Fault Tree Analysis (FTA)** and **DFMEA workflow pipelines** using **Python** and **Jenkins CI/CD** to correlate hazard severity rankings with design controls, test coverage, and corrective action status; enabled real-time **Tableau executive dashboards** displaying patient safety metrics, closing **126 critical and major findings** across 3 FDA pre-market submissions.
- Spearheaded cross-functional **independent failure investigations** and **hazard analysis workshops** with clinical, hardware, and software engineering teams; identified **18 latent patient safety risks** in continuation-engineering releases using **PHA, HAL, and DFMEA techniques**, delivered documented preventive controls, and achieved **zero critical non-conformances** in subsequent design review cycles.
- Developed risk-benefit decision-support analytics using **Power BI DAX measures** and **statistical hypothesis testing (R)** to quantify patient harm scenarios, mitigation effectiveness, and competitive risk positioning; delivered **quarterly safety and reliability briefings** to executive and regulatory stakeholders, enabling data-driven design prioritization that reduced post-market safety complaints by **23%**.

SWIFT Payments Operations Dashboard & ISO 14971 Risk Monitoring (HCL Technologies)

- Engineered a real-time **ISO 14971 risk-aware SWIFT messaging operations dashboard** combining payment transaction visibility, hazard alert escalation, and regulatory compliance tracking for a Tier-1 banking client; integrated **Splunk log aggregation, Dynatrace APM, and AWS CloudWatch metrics** into **Power BI interactive reports**, reducing mean time to hazard detection by **52%** and eliminating **4 post-incident root-cause blind spots**.
- Applied **Preliminary Hazard Analysis (PHA)** and **Design Failure Mode Analysis (DFMEA)** methodologies to map SWIFT **MT/MX message flows, ISO 20022 payment routing logic, and SEPA cross-border transaction paths** to patient/customer safety risks; documented **34 hazardous scenarios** spanning message

corruption, latency exceedance, and regulatory (GDPR, PSD2) violations, delivered design mitigations in **3 sprint cycles**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)